

IS 733 - Lab 11

Part I: Distributional Hypothesis

Please use the Distributional Hypothesis Concept to find the solution, show your work step by step

Example 1

1. A piece of _____ is on the plate.
2. Everyone enjoys eating _____.
3. You can cut _____ with a knife.
4. We make _____ from milk.

Potential Answers: "cheese," "cake," or "butter"

ANSWER:

CHEESE

Explanation: “Cheese” fits all sentences. It is made from milk, can be served in pieces on a plate, is commonly eaten, and can be cut with a knife. "Cake" and "butter" don't fit all four contexts as well as "cheese" does.

Example 2

1. The _____ is parked in the driveway.
2. He bought a new _____ for his birthday.
3. _____ can drive really fast.
4. People often wash their _____ on the weekends.

Potential Answers: "car," "truck," or "motorcycle"

ANSWER:

CAR

Explanation: All three options are vehicles, but “car” is the most commonly associated with parking in a driveway, birthday gifts, driving fast, and being washed. It is the most general and fits all sentence contexts well.

Example 3

1. I read an interesting _____ last night.
2. Many people enjoy a good _____ before bed.

3. _____ often has chapters and a cover.
4. You can borrow a _____ from the library.

Potential Answers: "book," "novel," or "story" – these terms appear in contexts related to reading and library usage.

ANSWER:

BOOK

Explanation: “Book” fits all four sentences clearly. It can be borrowed from a library, has chapters and a cover, and is commonly read at night. "Novel" is a specific kind of book, and "story" may not always be a physical object.

Part II: Practice the NLP model to classify data stories

Version	Experiment	Model	Training Data			
			Combined	Zero Shot	One Shot	Two Shot
LLM Generated Stories	Cross Validation	Logistic Regression	0.93	0.85	0.79	0.86
		Naive Bayes	0.92	0.79	0.78	0.82
		SVM	0.94	0.82	0.79	0.83
	Leave One Plot Out	Logistic Regression	0.95	0.94	0.92	0.95
		Naive Bayes	0.95	0.94	0.93	0.95
		SVM	0.95	0.94	0.94	0.95
Student Generated Stories	Train/Validate	Logistic Regression	0.77	0.68	0.69	0.78
		Naive Bayes	0.77	0.67	0.70	0.77
		SVM	0.76	0.67	0.68	0.77

Figure 6: ML model performance (Area Under Curve) in discriminat- ing whether a given narrative sentence is “show” versus “tell”

ANSWER:

1)A SHORT PARAGRAPH SUMMARIZING YOUR FINDINGS:

In this lab, I implemented a text classification pipeline to distinguish between “Show” (Level 1) and “Tell” (Levels 2/3) narrative sentences using natural language processing and machine learning. After preprocessing the text using lowercase conversion, stop word removal, and lemmatization (with SpaCy), I transformed the text using TF-IDF vectorization. I trained four models—Logistic Regression, SVM, Naive Bayes, and Random Forest—and evaluated their performance using 5-fold cross-validation. Among them, the Support Vector Machine (SVM) achieved the highest accuracy of **86.86%**, showing strong performance in identifying the narrative style based on textual patterns.

2) A **GPT STATEMENT:**

Large language models like GPT can assist in narrative classification tasks by generating additional labeled examples, improving text representations using embeddings, or serving as zero-shot classifiers without requiring training data. In this lab, GPT could be used to enhance feature quality beyond TF-IDF by capturing deeper semantic context, or to automate initial labeling of data stories based on prompt instructions. While traditional models like SVM performed well, integrating GPT could further improve classification performance and reduce manual effort.

3) THE **PYTHON NOTEBOOK FILE:**

Lab11_Text_classification.ipynb